Unit 2.1 Printables

DNA to Proteins (HS-LS1-1) · 8-day version: Siamese Cats

Day	Print	Copies
1	Day 1 worksheet (data tables and questions)	1 per student
2	Day 2 worksheet	1 per student
3	Gummy DNA lab sheet; strand mat and base key; code table	1 per student; 1 mat set per pair
4	Code table and card sort (cards A to F, three category mats)	1 set per group
5	Day 5 worksheet	1 per student
6	Day 6 case data worksheet	1 per student
7	Vocabulary and Write It page	1 per student
All	Student notebook spreads (separate file: Unit_2.1_Student_Notebook.pdf)	1 per lesson, or have students copy the layout

Lab materials checklist (Day 3, per pair)

- 2 licorice strands (6 to 8 inches)
- 24 candy bases: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G), plus a few spare
- Toothpicks, a paper plate, the printed lab sheet and code table
- No food? Use colored beads and pipe cleaners, or colored paper circles

Tape-in tables are sized to fit half of a notebook page. Cut on the dashed lines.

Day 1

Day 1: Why are a Siamese cat's ears, paws, and tail dark?

Notice and wonder

Look at the photos of an adult Siamese cat and a newborn Siamese kitten.

I notice: _____ I notice: _____

I wonder: _____ I wonder: _____

Data 1: Temperature and color

Body part of an adult Siamese cat	Skin temperature (°C)	Fur color
Back (trunk)	36	Cream
Belly	36	Cream
Upper leg	34	Cream
Ear	28	Dark brown
Nose and face mask	30	Dark brown
Paw	28	Dark brown
Tail tip	27	Dark brown

Simplified class data set. Real temperatures vary.

1. What pattern do you see between skin temperature and fur color?
2. A Siamese kitten is born almost white, and the dark parts appear in the first weeks. Use the data to suggest why.

Data 2: Litters of kittens

Litter	Mother	Father	Dark-pointed (Siamese pattern) kittens	Solid-colored kittens
1	Siamese pattern	Siamese pattern	5	0
2	Siamese pattern	Siamese pattern	4	0
3	Siamese pattern	Solid	0	6
4	Solid	Solid	1	4
5	Solid	Solid	2	3
6	Siamese pattern	Solid	3	2
7	Solid	Solid	1	5

Simplified class data set modeled on real litter patterns.

3. Describe the pattern for each type of pair: Siamese × Siamese, Siamese × solid, and solid × solid.
4. Is the Siamese pattern passed from parents to kittens? What is your evidence? Does a kitten need something from one parent or from both?

Day 1

Tape-in tables

Cut out and tape into your notebook.

Data 1: Temperature and fur color

Body part of an adult Siamese cat	Skin temperature (°C)	Fur color
Back (trunk)	36	Cream
Belly	36	Cream
Upper leg	34	Cream
Ear	28	Dark brown
Nose and face mask	30	Dark brown
Paw	28	Dark brown
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Data 2: Litters of kittens

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3	Siamese pattern	Solid	0	6
4	Solid	Solid	1	4
5	Solid	Solid	2	3
6	Siamese pattern	Solid	3	2
7	Solid	Solid	1	5

Simplified class data set modeled on real litter patterns.

Key ideas

- **DNA** in every cell stores instructions.
- A **gene** is a section of DNA with the instructions for one **protein**.
- **Proteins** do jobs for the cell.
- A kitten gets one set of chromosomes from each parent.

Day 1

Key ideas and practice

- **DNA** in every cell stores instructions.
- A **gene** is a section of DNA with the instructions for one **protein**.
- **Proteins** do jobs for the cell.
- A kitten gets one set of chromosomes (long DNA molecules with many genes) from each parent.

Practice

5. The molecule that stores the instructions in a cat's cells is _____.
6. A section of DNA with the instructions for one protein is a _____. Its product, which does a job for the cell, is a _____.
7. How many sets of chromosomes does a kitten get from each parent?
8. **Initial model.** Draw and label a model of what is inside a Siamese cat's cells that leads to dark ears, paws, and tail. Use arrows. We will revise it on Day 7.

Day 2

Day 2: What is DNA made of?

Key ideas

- DNA is a **double helix**, like a twisted ladder.
- It is built from **nucleotides**. Each nucleotide has a **sugar**, a **phosphate**, and a **base**.
- The sugars and phosphates make the sides (the backbone). The bases make the rungs.
- There are four bases: **A, T, C, G**. A pairs with T. C pairs with G.
- **The order of the bases is the message.**

Part A: One nucleotide

Draw one nucleotide: a pentagon (sugar), a circle (phosphate), and a rectangle (base). Label each part and color the base.

Part B: Write the partner strand

1. A T G C C T → _____
2. T T T G A C → _____
3. C T G G G A → _____

Part C: Questions

1. What shape is DNA?
2. What makes up the sides of the ladder? What makes up the rungs?
3. Which bases pair together?
4. What makes one gene different from another?
5. **Illustration only.** A short stretch of a gene in two cats (made-up letters that show the idea):
   Solid cat: A T G G G A C T G T T T
   Siamese cat: A T G A G A C T G T T T
   How many letters differ? Where?

Day 3

Gummy DNA Lab: Build, Copy, and Read a Gene

Name: _____ **Partner:** _____

Question: How does a cell's copy of a gene become a chain of amino acids, and what happens if one letter of DNA changes?

You will build a short model of the start of a cat's gene for a pigment-making protein. (The letters are an illustration, not the real gene.)

Materials (per pair)

- 2 licorice strands (the backbone; the sugars and phosphates)
- Candy “bases”: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G) gummies, plus 2 extra of each color
- Toothpicks (to attach bases to the backbone and to hold pairs together)
- Paper plate or tray, this sheet, the code table

Safety: Keep materials clean. Do not eat the lab materials until your teacher says you may. Toothpicks are sharp. Tell your teacher about any food allergy. If candy is not used, use colored beads or paper circles.

Base key and code table

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Part A: Build the template strand (6 min)

This is the strand of the gene that the cell uses as a template. Attach the bases to one licorice strand in order from left to right:

T A C C C T G A C A A A

Write the color order you built: _____

Part B: Build the partner strand and make the ladder (8 min)

Build the partner strand on the second licorice strand by pairing each base (A with T, C with G). Write the partner sequence you built:

_ _ _ _ _ _ _ _ _ _ _ _

1. Join the two strands with toothpicks between pairs. Gently twist the ladder. Have your teacher check it before you continue. ☐ Checked
2. Which strand carries the “coding” letters (the same as the cell's message copy)? How do you know?

Day 3

Part C: Copy the message, then read it (10 min)

The cell does not use the DNA directly to make a protein. It **copies the gene's message** from the template strand, and then the **copy is read** to link amino acids. The DNA stays safe.

1. Pull the two strands apart. Keep the template strand. Build a *new* strand by pairing with the template. This is the **copy**. Write its sequence: ___ ___ ___ ___
2. Read the copy three letters at a time with the code table. Write the amino acid chain: _____ – _____ – _____ – _____
3. Line up one bead or paper tag per amino acid on the plate in order. This chain would fold to make a protein.
4. Explain the path in your own words, starting with the DNA:
   _____

Part D: Change one letter (7 min)

1. On your template strand, replace the 4th base (the first C of the second triplet) with a T (swap a green gummy for a yellow one). The template now reads **TAC TCT GAC AAA**.
2. Build the new copy by pairing with the changed template. Sequence: ___ ___ ___ ___
3. Read it with the code table: _____ – _____ – _____ – _____
4. Compare with Part C. How many amino acids changed? Which one?
5. Predict: Could a protein built from this changed chain work differently? Explain.

Connect to the Siamese cat

1. In a Siamese cat, a one-letter change in the gene for a pigment-making protein swaps one amino acid. In your own words, what did your one-letter change do to the chain in the model?
2. Name two ways your gummy model is different from real DNA.

Clean up

Put toothpicks and candy in the trash or the labeled container. Wipe your table. Wash your hands.

Day 3

Gummy DNA mats

Slide the licorice strand along the mat. One base per box, three boxes per amino acid.

Base key

Red = A Yellow = T Green = C Orange = G

Part A and B: template strand (T A C C C T G A C A A A) and partner strand

Template

Partner

Part C: the copy

Amino acids: _____ – _____ – _____ – _____

Part D: changed template (T A C T C T G A C A A A) and its copy

Amino acids: _____ – _____ – _____ – _____

Code table (tape in)

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Day 4

Card sort

Cut on the dashed lines. Sort into three categories and write your reason on the back.

A

In a Siamese cat, one letter changes in the gene for a pigment-making protein: GGA becomes AGA.

B

A gene's triplet changes from GAG to GAA.

C

A gene's triplet changes from CTG to CCG.

D

One letter changes in a stretch of DNA that has no gene and no switch.

E

One letter changes in a switch next to a gene so the gene can never be turned on, and its protein is not made.

F

In a human gene, GAG becomes GTG.

Day 4

Category mats and code table

Category 1	Category 2	Category 3
Changes an amino acid in the chain	Changes whether or when the protein is made	No noticeable change

Code table

DNA triplet	Amino acid	DNA triplet	Amino acid
GGA	Gly	AGA	Arg
GAG	Glu	GTG	Val
GAA	Glu	CTG	Leu
CCG	Pro		

Questions

1. Write the cards in each category and your reasons.
2. A student says, "Every change in DNA changes how a cell works." Use the cards to respond.
3. Card A changes one amino acid. Predict what could happen to the protein and to the fur color. (Hint: think about temperature.)

Day 5

Day 5: Same DNA, different jobs

Key ideas

- Every cell has the same DNA, including all the genes.
- Each kind of cell uses a different set of genes, makes different proteins, and so has a different job.

Part A: Two kinds of cells

Cell type	Genes in the cell	Proteins made	Job
P	1, 2, 3, 4	1 and 2	Carries signals
Q	1, 2, 3, 4	3 and 4	Squeezes to move the body

Do P and Q have the same DNA? How do you know? Why do they have different jobs?

Part B: Cells in a Siamese cat's ear

Cell type	Has the gene for the pigment-making protein?	Makes the pigment-making protein? (X)	Job
Pigment-making cell			Makes dark color in fur
Skin cell			Covers and protects
Nerve cell			Carries signals

Part C: Where does the changed protein work?

In a Siamese cat, the pigment-making protein has one amino acid changed. It works only in skin cooler than about 33 °C (a simplified number). Complete the table using the temperatures.

Body part	Skin temperature (°C)	Does the protein work? (yes or no)	Fur color (dark or cream)
Ear	28		
Nose	30		
Paw	28		
Tail tip	27		
Upper leg	34		
Back	36		
Belly	36		

1. Do all of the cat's cells have the gene for the pigment-making protein?
2. Which cells make the protein?
3. Why are only the ears, nose, paws, and tail dark?
4. **Explain.** Use the words DNA, gene, protein, and job to explain why a Siamese cat has dark points.

Day 6

Day 6: Case data

Case 1: Siamese and solid cats

	Solid cat	Siamese cat
Gene for the pigment-making protein	Usual sequence	One letter different
Amino acid chain	Usual chain	One amino acid different
Where the protein works	At every body temperature	Only in skin cooler than about 33 °C
Fur color	Dark all over	Dark ears, nose, paws, tail; cream body
Temperature of ear / back	28 / 36 °C	28 / 36 °C

Fill in the chain: One letter in the _____ changes → one _____ in the protein is different → the protein works only _____ → pigment is made only on the _____.

Case 2: A human gene

Person	Gene	Protein	Symptoms
1	Normal	Works	None
2	One base changed	One amino acid different; works poorly	Yes

Fill in the chain: One base in the _____ changes → one _____ in the chain is different → the _____ works poorly → _____.

1. How are the two cases alike?
2. How are they different?
3. Why does a change in DNA change what a protein can do?
4. In Case 2, why does a different amino acid cause symptoms?
5. What do the Day 1 litter data tell you about the changed gene in Case 1?
6. **Write.** Claim: What caused the Siamese cat's dark points? Evidence: use at least two numbers from the data. Reasoning: explain how a one-letter DNA change leads to dark color only on the cooler body parts.

Day 7

Vocabulary and Write It

Vocabulary

Word bank: DNA, nucleotide, gene, base sequence, amino acid, protein

1. The molecule that stores the instructions for building proteins is _____.
2. One building block of DNA, with a sugar, a phosphate, and a base, is a _____.
3. A section of DNA with the instructions for one protein is a _____.
4. The order of A, T, C, and G along a strand of DNA is its _____.
5. A chain of _____ is folded to make a protein.
6. A _____ does jobs for the cell.

Write It: two Siamese kittens

	Kitten raised warm	Kitten raised cool
Ear skin temperature (°C)	34	26
Ear fur color	Light tan	Dark brown
Back skin temperature (°C)	36	35
Back fur color	Cream	Cream

Claim: why are the cool-room kitten's ears darker? **Evidence:** at least two numbers. **Reasoning:** DNA, gene, protein, and what the amino acid change does.